

Sweep + Sketch Tutorial

Building helical springs, arched handles, bezier curves, custom hooks — anything that's a 2D profile dragged along a 3D path.

Sweep is ForgeSlicer's most flexible primitive. Combined with 2D sketches, it lets you draw an arbitrary cross-section and then extrude it along any path — helix, arc, bezier, or your own hand-drawn polyline.

1. What is a sweep?

A **sweep** is a 2D shape (the *profile*) translated along a 3D curve (the *path*). Imagine pulling a cookie cutter through the air: the cookie cutter is the profile, the trajectory of your hand is the path, and the resulting solid is the sweep.

ForgeSlicer's sweep primitive is built on this exact idea, with four extras: **twist** (the profile rotates as it travels), **samples** (how many cross-sections to build), and the ability to source either the profile or the path — or both — from a hand-drawn sketch.

Two builders ship in the box

Field	Options
profile.kind	circle · square · rect · star · sketch (your own 2D drawing)
path.kind	helix · arc · bezier · sketch3d (your own 3D polyline) · ref (an existing scene curve)

2. Anatomy of a sweep object

Parameter	Effect	Default
samples	How many cross-sections along the path. Higher = smoother but more triangles.	96
twistDeg	Total rotation of the profile across the entire path (degrees). 0 = no twist; 360 = one full revolution.	0
profile.kind	Shape of the cross-section.	circle
profile.r / .w / .h / .points	Profile dimensions, kind-dependent. Sketch profile carries a flat <i>points</i> array.	—
path.kind	Trajectory shape.	helix
path.r / .pitch / .turns / .points	Path geometry, kind-dependent. Sketch3D carries a 3D <i>points</i> array.	—

Triangle budget: samples=96 produces about samples × profile-vertex-count triangles. A circle profile (r=2, segments=16) at samples=96 makes ~1.5 k triangles. A star profile with 10 points easily doubles that. Drop samples to 48 before slicing if you don't see micro-detail in the result.

3. Path kinds in depth

Helix

A spring or screw thread. Parameters: **r** (helix radius), **pitch** (mm rise per turn), **turns** (number of complete rotations). Total height = pitch × turns. The path starts at (r, 0, 0) and spirals up the Y axis.

Common helix recipes:

- **Coil spring** — circle profile r=1.5, helix r=10, pitch=4, turns=8. Result: a classic spring 32 mm tall.
- **Threaded bolt** — star profile (5 points, inner r=2, outer r=2.5), helix r=2.5, pitch=0.8, turns=37. Matches an M5 thread spec.
- **Vase rim** — rect profile 2×0.5, helix r=15, pitch=0.2 (very tight), turns=200. A barely-perceptible spiral lip around a cylinder.

Arc

A circular arc in a single plane (XY by default). Parameters: **r** (arc radius), **startDeg / endDeg** (angle span in degrees). Use this for handles, hooks, brackets, anything bent at a consistent radius.

Bezier

A cubic bezier curve defined by four control points (**p0**, **p1**, **p2**, **p3**). The curve starts at p0, ends at p3, and the middle two points pull the curve in their direction without actually touching it. The Inspector exposes all four as XYZ vec3 inputs.

Bezier tip: for a smooth handle, place p0 and p3 at the two endpoints, then drag p1 and p2 to set the curve's *tangent* direction at each end. The strength of the pull is roughly the distance from the endpoint to its companion control point.

Sketch3D — your own polyline

An array of explicit (x, y, z) points connected by a smooth CatmullRom interpolation. Use the Sketch tool (Toolbar → Sketch · Pencil) to draw the path in 2D, then promote it via right-click → **Use sketch as Sweep path (3D)**. The Inspector then exposes a **Rise (mm)** field that lifts the polyline linearly along Y — turn a flat doodle into a sloping ramp with one number.

Ref — an existing scene curve

Sometimes you want the sweep to follow the edge of another part. Set *path.kind* to **ref** and pick a target object — the sweep uses that object's first-component centerline as its path. Lets you build trim seals, gaskets, or wire-wraps that hug another part's geometry.

4. Profile kinds

Kind	Parameters	Use case
circle	r (radius), segments (default 16)	Wire, tubes, springs, generic rounded extrusions.
square	w (side length)	Rectangular trim, square handles.
rect	w × h	Flat ribbon, belt-like sweeps.
star	points (count), innerR , outerR	Threads, gear-tooth profiles, decorative star-extrusions.
sketch	points — array of [x, y] pairs	Truly custom — draw the profile in Sketch mode, right-click → Use sketch as Sweep profile , then tune in the Inspector.

5. Hand-drawn profiles & paths — the Sketch tool

Ridges and stars are great, but real designs need shapes that don't fit any preset. The Sketch tool gives you a 2D drawing canvas that can become either a profile or a path:

Drawing a sketch

- 1 Click the **Sketch** button in the top toolbar (or press **K**). The viewport switches to a 2D overlay with a grid.
- 2 Choose a tool: **Pencil** for free-form polylines, **Line** for straight segments, **Rect** for closed rectangles, **Circle** for closed loops, or **Spline** for smooth curves between control points.
- 3 Snap-to-grid is on by default (1 mm). Hold Shift to disable while you drag.
- 4 When you're happy, click **Commit**. The sketch becomes a scene object (visible in the outliner as *Sketch 1* etc.).

Promoting a sketch to a sweep

Right-click your committed sketch in the viewport. Two new entries appear in the context menu:

- **Use sketch as Sweep profile** — creates a new sweep object with *profile.kind* = **sketch**, defaulting to a helix path. Edit the path in the Inspector afterwards.
- **Use sketch as Sweep path (3D)** — creates a new sweep with *path.kind* = **sketch3d**, defaulting to a small circular profile. A new **Rise (mm)** field appears in the Inspector that lifts the polyline along Y.

The sketch is preserved. Promoting a sketch to a sweep duplicates its points into the new sweep object — the original sketch stays in the scene so you can keep iterating, then re-link the sweep to a new version via the Inspector's picker.

6. Walkthrough — a custom hook

Goal: a hook with a flat tang, a curved shaft, and a tapered tip — the kind you'd 3D-print to hang a thing on a wall.

- 1 Click **Sketch** (or press **K**). Draw the hook profile in 2D: a flat-bottomed shape ~6 mm wide × 4 mm tall, with a tapered tip on one side. Click **Commit**.
- 2 Right-click the sketch → **Use sketch as Sweep profile**. A new sweep object appears with the profile set, defaulting to a helix path.
- 3 In the Inspector, change **path.kind** from helix to **arc**. Set **r** = 25 mm, **startDeg** = 0, **endDeg** = 240. The sweep now follows a 240° arc.
- 4 Set **samples** to 128 for a smooth result. Tune **twistDeg** to 0 (the hook profile shouldn't twist along the arc).
- 5 If the hook is the wrong way up, rotate the whole sweep object -90° on Z via the Inspector to stand it upright.
- 6 Slice it. The G-code preview should show a smooth curved hook with your custom profile preserved end-to-end.

7. Twist + samples — what they really do

twistDeg is the *total* rotation applied to the profile across the ENTIRE path, regardless of length. A 360° twist on a 10 mm path is one full revolution per 10 mm; the same 360° on a 1000 mm path is one revolution per metre. Set twist *after* the path is dialled in.

samples is how many cross-sections to build. The renderer interpolates between them — too few and the sweep looks faceted; too many and the triangle count blows up. Rules of thumb:

- **Short paths (<30 mm), simple profile.** 48 samples plenty.
- **Medium paths (30–150 mm), or any twist.** 96–128 samples — the default 96 is correct.
- **Long paths (>150 mm) or fine helix (turns > 10).** 192–256 samples to keep the spiral smooth.
- **If frame rate dips while editing.** Drop samples temporarily; bump back up before slicing.

8. Print tips for swept geometry

- **Overhangs are inevitable on curved paths.** Anything past 45° from vertical needs supports. Use OrcaSlicer's tree-support mode for organic-shaped sweeps — way fewer support scars on a curved part than grid supports.
- **Helical springs print best lying down.** Standing them up is theoretically possible but the small thread cross-section fails at the layer-bonding interface. Lay flat, use one wall, 100% infill on the swept profile only.
- **Long arcs need orientation thought.** A hook printed point-up needs supports under the tip. Same hook flat-on-bed prints in one shot with no supports.
- **Tiny twist + tiny samples = facets you can FEEL.** If you twist 360° over 50 samples, every 7.2° you get a visible flat. Bump samples or reduce twist.

9. Troubleshooting

Symptom	Likely cause	Fix
Sweep renders as a flat ring	Path has only one point (e.g., a sketch that was never closed).	Open the sketch picker in the Inspector and pick a valid 3+ point sketch.

Sweep has self-intersection / non-manifold STL	Path turns more sharply than the profile's diameter.	Either increase samples (smoother turn) or shrink the profile.
Twist looks wrong in one direction	twistDeg can be negative — sign controls handedness.	Set twistDeg to −value for opposite handedness.
'Use sketch as Sweep profile' menu item missing	Right-click target is not a single sketch object.	Select exactly one sketch in the outliner first, then right-click.
Bezier curve goes the wrong way	Control points 1 and 2 are swapped — the curve is mirrored.	Edit the p1, p2 fields in the Inspector — bezier is order-sensitive.

10. Quick reference

- Keyboard shortcuts:** K opens Sketch mode · Esc cancels the current draw · Enter commits a polyline.
- Source:** sweep math lives in frontend/src/lib/sweepGeometry.js (buildSweepGeometry + SWEEP_DEFAULTS).
- Inspector:** frontend/src/components/SweepInspectorBlock.jsx renders the per-kind controls.
- Sketch overlay:** frontend/src/components/SketchOverlay.jsx.
- Tests:** frontend/tests/sweep-geometry.mjs, frontend/tests/sketch-to-sweep.mjs.

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